Device Compliance Policies

Implementation Solution Guide

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Executive Summary — Device Compliance Policies (Microsoft Intune)

White-label · Intune Solutions

An unmanaged device with no encryption, a weak PIN, or an outdated OS is a breach waiting to happen — and most organizations have no automated way to *prove* a device is healthy before it touches company data. **Device Compliance Policies** in Microsoft Intune define the health rules every device must meet, continuously evaluate them, and feed that status to Conditional Access so only compliant devices get in.

This is a complete, **deploy-it-today** package for implementing Device Compliance Policies with Microsoft Intune, written for an administrator with no prior compliance-policy experience.

What you get

- A plain-language **Solution Overview** of how compliance evaluation and enforcement work.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for local health discovery, compliance validation, tenant reporting, remediation, and conflict checking.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

A Windows compliance policy enforcing **encryption (BitLocker), Secure Boot, minimum OS, password rules, antivirus/firewall**, with automated **actions for noncompliance** (notify, then mark noncompliant) — ready to gate access via Conditional Access.

The standard deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Provable device health, automated enforcement, and a compliance signal that powers Zero Trust access control — deployable by a single administrator and validated before fleet rollout.

1. Solution Overview — Device Compliance Policies

What Device Compliance Policies are

A device compliance policy is a set of health rules (encryption, OS version, password, antivirus, threat level, and more) that Microsoft Intune evaluates on each enrolled device. The device is marked **Compliant** or **Not compliant**, and that status becomes a signal Conditional Access can use to allow or block access to Microsoft 365 and other apps.

How it works

1. You define a compliance policy per platform with the rules a device must satisfy.
2. Intune evaluates the device against the rules on enrollment and on a recurring schedule.
3. The result (Compliant / Not compliant) is stored against the device in Entra ID.
4. **Actions for noncompliance** run on a schedule — notify the user, then mark the device noncompliant after a grace period.
5. **Conditional Access** can require "compliant device" to grant access, turning compliance into enforcement.

What you can require (Windows)

- **Device health** — BitLocker, Secure Boot, Code integrity.
- **Device properties** — minimum/maximum OS version.
- **System security** — password required, length, complexity; encryption of data storage.
- **Microsoft Defender** — antivirus, antispyware, real-time protection, firewall on.
- **Defender for Endpoint** — require a machine risk score at or below a threshold (if integrated).

Benefits

- **Provable device health** before access is granted.
- **Automated enforcement** — notify and mark noncompliant without manual effort.
- **Zero Trust foundation** — the compliance signal powers Conditional Access.
- **Visibility** — per-device and tenant-wide compliance reporting.

Limitations (be honest with buyers)

- Compliance only **enforces** access when paired with Conditional Access (a companion solution).
- Threat-level rules require Microsoft Defender for Endpoint integration.
- A grace period means a freshly noncompliant device isn't blocked instantly (by design).
- Devices without any assigned policy are treated per your tenant default — set it deliberately.

The standard deployment model

> ***Set the no-policy default → create the compliance policy → configure actions → assign to a pilot group → (optionally) require compliance in Conditional Access.***

repeatable pattern: **Assess → Configure → Pilot → Validate → Expand.**

2. Requirements Guide — Device Compliance Policies

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Intune (Plan 1) for compliance; Microsoft Entra ID P1 for Conditional Access enforcement. **Why:** Intune evaluates compliance; Entra P1 lets Conditional Access require a compliant device. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium (both include Intune + Entra P1).

2.2 Device enrollment

Description: Targeted devices are enrolled in Intune. **Why:** Compliance is only evaluated on enrolled devices. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

2.3 Device health prerequisites

Description: TPM + UEFI Secure Boot for the health rules (BitLocker, Secure Boot). **Why:** Health attestation rules require TPM/Secure Boot capable hardware. **Verify:** `Get-Tpm`; `Confirm-SecureBootUEFI`; or run `DCP-Discovery.ps1`. **Implement:** Enable TPM and Secure Boot in UEFI; this solution can be paired with the BitLocker solution to enforce encryption.

2.4 Microsoft Defender (for AV/firewall rules)

Description: Microsoft Defender Antivirus + Windows Firewall active (or a managed AV). **Why:** Compliance can require real-time protection and firewall on. **Verify:** `Get-MpComputerStatus`; `Get-NetFirewallProfile`. **Implement:** Ensure Defender AV is enabled (default) and firewall profiles are on.

2.5 Defender for Endpoint (optional — threat level rules)

Description: Defender for Endpoint connector in Intune. **Why:** Required only if you enforce a machine risk score. **Verify:** Intune → **Endpoint security** → **Microsoft Defender for Endpoint** connection status = Active. **Implement:** Connect Defender for Endpoint (see the Defender integration solution).

2.6 Conditional Access (for enforcement)

Description: A Conditional Access policy requiring a compliant device. **Why:** Compliance reports status; Conditional Access turns it into access enforcement. **Verify:** Entra → **Protection** → **Conditional Access**. **Implement:** Pair with the Conditional Access Integration solution.

Checklist

- Intune + Entra P1 licensing assigned
- Devices enrolled

- TPM + Secure Boot for health rules — `DCP-Discovery.ps1`
- Defender AV + firewall active
- (Optional) Defender for Endpoint connected for threat-level rules
- Pilot Entra group created
- Tenant "no compliance policy" default decided

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Device Compliance Policies

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *A Windows compliance policy with health + password + Defender rules, actions for noncompliance, assigned to a pilot group — ready for Conditional Access enforcement.*

Step 1 — Set the no-policy default

1. Go to **Devices** → **Compliance** → **Compliance settings**.
2. Set **Mark devices with no compliance policy assigned as** to **Not compliant**. **Save**.

Step 2 — Create the compliance policy

1. Go to **Devices** → **Compliance** → **Create policy**.
2. **Platform:** Windows 10 and later. Click **Create**.
3. **Name:** `Compliance - Windows - Pilot`. Click **Next**.

Step 3 — Configure the rules

On **Compliance settings**, set:

1. **Device Health** → Require BitLocker = **Require**; Require Secure Boot = **Require**; Code integrity = **Require**.
2. **Device Properties** → Minimum OS version = your baseline (e.g. `10.0.19044`).
3. **System Security** → Require a password = **Require**; Minimum password length = **8**; Password type = **Alphanumeric** (or Device default); Encryption of data storage = **Require**.
4. **Microsoft Defender Antivirus** = **Require**; **Real-time protection** = **Require**; **Firewall** = **Require**.
5. (Optional) **Defender for Endpoint** → Require the device to be at or under machine risk score = **Medium**.

6. Click **Next**.

Step 4 — Configure actions for noncompliance

1. On **Actions for noncompliance**, keep **Mark device noncompliant** and set schedule to **1 day** (grace period).
2. Click **Add** → **Send email to end user** → choose/create a notification → schedule **immediately**.
3. (Optional) Add **Remotely lock the noncompliant device** for higher-risk scopes. Click **Next**.

Step 5 — Assign to the pilot group

1. On **Assignments** → **Included groups** → **Add groups** → select `Compliance-Pilot`. **Next**.

Step 6 — Review and create

1. On **Review + create**, confirm the rules and actions. Click **Create**.

Step 7 — (Recommended) require compliance in Conditional Access

1. Pair with the **Conditional Access Integration** solution: create a policy requiring **Require device to be marked as compliant** for the same users. Start in **Report-only** mode.

Step 8 — Validate the pilot

1. On a pilot device, run `DCP-Validation.ps1` → confirms the device reports Compliant.
2. Run `DCP-TenantReport.ps1` → confirms compliance status across the pilot group.
3. Run `DCP-ConflictCheck.ps1` → flags conflicting policies. Complete the **Testing Guide**.

Step 9 — Expand to production

1. Add a production group to the policy **Assignments**; roll out in waves.
2. Move the Conditional Access policy from **Report-only** to **On** once compliance coverage is confirmed.
3. Re-validate each wave with `DCP-Validation.ps1` / `DCP-TenantReport.ps1`.

Order of operations

`No-policy default → Create policy → Rules → Actions → Assign → (CA require compliant) → Validate → Expand`

5. Testing Guide — Device Compliance Policies

Run after the pilot policy applies (Implementation Step 8) and after each production wave.

Test 1 — Policy applied

Do: Intune → **Devices** → **Compliance** → `Compliance - Windows - Pilot` → **Device status**. **Expected:** Pilot devices listed; evaluation completed (no **Error**).

Test 2 — Compliant device reports Compliant

Do: On a healthy pilot device, run `DCP-Validation.ps1`; check Intune → device → **Compliance**. **Expected:** Device is **Compliant**; all required rules pass.

Test 3 — Noncompliant device is caught

Do: On a test device, disable the firewall (or use a non-encrypted test VM) and force a sync. **Expected:** The device flips to **Not compliant** for the failing rule after evaluation.

Test 4 — Actions fire

Do: Leave a device noncompliant past the grace period. **Expected:** The user receives the noncompliance email; the device is marked **Not compliant** on schedule.

Test 5 — No policy conflicts

Do: Run `DCP-ConflictCheck.ps1`, and review Intune → device → **Device compliance** for conflicts. **Expected:** No conflicting compliance settings from multiple policies.

Test 6 — Conditional Access sees the signal (if configured)

Do: With a Report-only CA policy requiring compliant device, review **Entra** → **Sign-in logs** → **Conditional Access**. **Expected:** Compliant devices satisfy the control; noncompliant devices would be blocked (report-only shows "would block").

Test 7 — Tenant report

Do: Run `DCP-TenantReport.ps1`. **Expected:** A compliance breakdown (Compliant / Not compliant / In grace) for the pilot group.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–5 (and 6 if CA configured) pass for the pilot group
Wave pass	Tests 1–3, 7 pass before the next wave

6. Troubleshooting Guide — Device Compliance Policies

Use `DCP-Validation.ps1` / `DCP-Discovery.ps1` to confirm device state and `DCP-Remediation.ps1` for safe fixes.

A. Device shows Not compliant unexpectedly

Symptom	Cause	Remediation
Compliant device marked Not compliant	A required rule isn't met (firewall off, AV stale, OS below min)	Run `DCP-Discovery.ps1` to find the failing facet; remediate; sync
"Compliant" but no policy	No policy assigned + tenant default = Not compliant	Assign the policy to the device's group
Stuck "In grace period"	Grace period not elapsed	Wait for the scheduled action, or shorten the grace period

B. Evaluation / sync issues

Symptom	Cause	Remediation
Status not updating	Device hasn't checked in	Run `DCP-Remediation.ps1` (forces MDM sync); confirm device online
BitLocker/Secure Boot "not evaluated"	Health attestation not reporting	Ensure TPM + Secure Boot enabled; reboot; allow attestation to report

C. Policy conflicts

- Multiple compliance policies with conflicting settings → Intune uses the most restrictive, but mixed assignments cause confusion. Run `DCP-ConflictCheck.ps1`; consolidate to one authoritative policy per scope.
- A legacy GPO/another MDM setting the same security control can fight Intune — remove the duplicate.

D. Conditional Access interplay

- Users blocked unexpectedly → a CA policy requires compliant device but the device is Not compliant. Verify with **Entra** → **Sign-in logs**; fix the compliance failure or scope the CA policy.
- Always pilot CA in **Report-only** first to see impact before enforcing.

E. Defender for Endpoint threat-level rule

- Threat-level rule always failing → the Defender for Endpoint connector isn't Active, or the device isn't onboarded. Confirm the connector and onboarding (see the Defender integration solution).

F. Diagnostics quick reference

```
Get-MpComputerStatus                # Defender AV / real-time protection
Get-NetFirewallProfile              # firewall state
manage-bde -status                  # BitLocker
Confirm-SecureBootUEFI              # Secure Boot
```

Intune: **Devices** → **device** → **Device compliance** shows per-setting pass/fail.

G. When to escalate

If `DCP-Discovery.ps1` shows all facets healthy but the device still reports Not compliant after a sync and attestation cycle, collect the compliance evaluation details and open a Microsoft support case.

7. Best Practices — Device Compliance Policies

- **Always pilot first.** Apply to a small group and confirm healthy devices report Compliant before fleet rollout.
- **Set the no-policy default to Not compliant.** Otherwise unmanaged devices are treated as compliant — the opposite of what you want.
- **Use a grace period.** Mark noncompliant after ~1 day so a transient failure doesn't instantly block a user; notify the user immediately.
- **One authoritative policy per scope.** Avoid overlapping compliance policies with conflicting settings; consolidate.
- **Pair with Conditional Access in Report-only first.** See who *would* be blocked before enforcing, to avoid lockouts.
- **Match rules to reality.** Don't require BitLocker/Secure Boot on hardware that can't support it; segment by capability.
- **Notify users.** A clear email on noncompliance ("your device needs an update / encryption on") drives self-remediation.
- **Layer with the encryption/Defender solutions.** Compliance *checks* health; BitLocker and Defender solutions *deliver* it.
- **Review the tenant report regularly.** Track compliance percentage and chase persistent failures with `DCP-TenantReport.ps1`.
- **Document the baseline.** Record the required rules and rationale for audits and repeatability.

8. Deliverables Summary — Device Compliance Policies

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What compliance policies are, how evaluation/enforcement work, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered, click-by-click Intune steps

Document	Purpose
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`DCP-Discovery.ps1`	Local health facts: BitLocker, Secure Boot, Defender AV, firewall, OS version
`DCP-Validation.ps1`	Confirms the device reports Compliant and which rules pass/fail
`DCP-TenantReport.ps1`	Tenant-wide compliance status breakdown (Microsoft Graph)
`DCP-Remediation.ps1`	Safe remediation; forces sync and fixes common failures (firewall/AV)
`DCP-ConflictCheck.ps1`	Flags conflicting/overlapping compliance policy assignments (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Provable device health and automated enforcement that powers Zero Trust Conditional Access.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

DCP-ConflictCheck.ps1

DCP-Discovery.ps1

DCP-Remediation.ps1

DCP-TenantReport.ps1

DCP-Validation.ps1